

Chapter 6

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1 Compact Sets

Claim 6.1. *Let X be a finite topological space. Then X is compact.*

Proof. Since X is finite then 2^X is finite, thus $\mathcal{T}$ is finite because by definition it is a subset of the power set of X . Any open cover of X is a subset of $\mathcal{T}$ and thus is finite thus X is compact. $\square$

Claim 6.2. *Let C be a compact subset of $\mathbb{R}$. Then C has a maximum point $m \in C$ such that $\forall c \in C$ $c \leq m$.*

Proof. Assume the contrary that C doesn't have a maximum m , thus $\forall x \in C, \exists y_x \in C \ni y_x > x$. Create the open cover of C , $U = \{(-\infty, y_x)\}_{x \in C}$. Any finite subcollection of U is defined by a finite subset of C , $D = \{x_1, \dots, x_n\} \subset C$. $\exists d \in D \ni \exists y \in C \setminus D \ni y > d$. Thus y is not in the union of the finite subcollection of U , thus the finite subcollection of U doesn't create a subcover of C . However this is a contradiction with the fact C compact. Thus C must have a maximum m such that $\forall c \in C \ni m \geq c$. $\square$

Claim 6.3. *If X has a compact space, then any infinite subset of X has a limit point*

Proof. Assume the contrary that exists a infinite subset A of X which has no limit points. Thus $\forall x \in X, \exists U_x \in \mathcal{T} \ni x \in U_x$ but $(U_x - \{x\}) \cap A = \emptyset$. Create the open cover $\mathcal{U} = \{U_x\}_{x \in X}$. Any finite subcollection of $\mathcal{U}$ will be given by a finite subset of X , $S = \{x_1, \dots, x_n\} \subset X$, where the open sets are U_{x_i} for all $i \in [1, \dots, n]$. However since A is infinite and S is finite, $S \cap A$ is finite. Since $\forall x \in X, (U_x - \{x\}) \cap A = \emptyset, \forall x \in S \cap A, U_x \cap A = \{x\}$. Thus $\forall x \in A - S \cap A, x \notin U_y$ for all $y \in S$. Thus the finite subcollection of $\mathcal{U}$ corresponding to S is not a cover. Thus the cover $\mathcal{U}$ has no finite subcover which contradicts X 's compactness. Thus by contradiction, all infinite subsets of X have a limit point. $\square$

Claim 6.4. *A space X is compact iff every collection of closed sets with the finite intersection property has a nonempty intersection*

Proof. $\implies$: Suppose X is compact. Let $\forall B \subseteq \mathcal{C}f$, such that B has the finite intersection property. $B = \{B_\alpha\}_{\alpha \in \lambda}$. Assume the contrary that $\bigcap_{\alpha \in \lambda} B_\alpha = \emptyset \implies \bigcup_{\alpha \in \lambda} B_\alpha^c = X$. Thus $B^c = \{B_\alpha^c\}_{\alpha \in \lambda}$ is a open cover of X . Thus there exists a finite subcover $D^c \subseteq B^c$ since X is compact. Thus $\bigcup_{\alpha \in [1, \dots, n]} D_\alpha^c = X \implies \bigcap_{\alpha \in [1, \dots, n]} D_\alpha = \emptyset$ thus there exists finite subcollection $D \subseteq B$ such that the intersection of the finite subcollection is empty, thus breaking the finite intersection property of B . Thus by contradiction, B must have a nonempty intersection.

$\Leftarrow$: Suppose X is a topological space such that every collection of closed sets with the finite intersection property have a nonempty intersection. Assume the contrary that X is not compact. Thus there exists a open cover of X , $\mathcal{U} = \{U_\alpha\}_{\alpha \in \lambda}$ which doesn't have a finite subcover of X . $\mathcal{U}^c = \{U_\alpha^c\}_{\alpha \in \lambda}$ is a collection of closed sets. Since $\bigcup_{\alpha \in \lambda} U_\alpha = X \implies \bigcap_{\alpha \in \lambda} U_\alpha^c = \emptyset$

Thus for all finite subcollections $\mathcal{U}_0 = \{U_n\}_{n \in [0, \dots, m]}$, $\bigcup_{n=1}^m U_n \neq X \implies \bigcap_{n=1}^m U_n^c \neq \emptyset$.

Thus $\mathcal{U}^c$ has the finite intersection property, thus its intersection must be nonempty which causes a contradiction. Thus X must be compact by contradiction. $\square$

Claim 6.5. *A space X is compact iff for any open set U in X and any collection of closed sets $\{K_\alpha\}_{\alpha \in \lambda}$ such that $\bigcap_{\alpha \in \lambda} K_\alpha \subseteq U$, there is a finite number of K_α 's whose intersection lies in U*

Proof. $\implies$: Suppose X is compact. Assume the contrary that $\exists U \in \mathcal{T}$ and a collection of closed sets $\mathcal{K} = \{K_\alpha\}_{\alpha \in \lambda} \ni \bigcap_{\alpha \in \lambda} K_\alpha \subseteq U$ but there doesn't exist a finite number of K_α 's whose intersection lies in U . $\mathcal{K}^c = \{K_\alpha^c\}_{\alpha \in \lambda}$ and $U^c \subseteq \bigcup_{\alpha \in \lambda} K_\alpha^c$. Thus $\mathcal{K}^c \cup U$ is a cover of X . Thus by definition of compact, $\exists \mathcal{K}_0^c = \{K_{\alpha_1}^c, \dots, K_{\alpha_n}^c\}$ a finite subcollection of $\mathcal{K}^c$ such that $\mathcal{K}_0^c \cup U$ is a finite cover of X . Thus $U^c \subseteq \bigcup_{i=1}^n K_{\alpha_i}^c \implies \bigcap_{i=1}^n K_{\alpha_i} \subseteq U$, thus there exists a finite number of K_α 's such that their intersection lies in U . Thus a contradiction occurs and the theorem is true.

$\Leftarrow$: Suppose for any open set U in X and any collection of closed sets $\{K_\alpha\}_{\alpha \in \lambda}$ such that $\bigcap_{\alpha \in \lambda} K_\alpha \subseteq U$, there is a finite number of K_α 's whose intersection lies in U . $\forall \mathcal{U} \subseteq \mathcal{T}$, $\mathcal{U} = \{U_\alpha\}_{\alpha \in \lambda}$ is a cover of X . $\forall V \in \mathcal{U}$, $V^c \subseteq \bigcup_{\alpha \in \lambda} U_\alpha^c$. Thus $\mathcal{U}^c = \{U_\alpha^c\}_{\alpha \in \lambda}$ is a collection of closed sets where $\bigcap_{\alpha \in \lambda} U_\alpha^c \subseteq V$. By the given, $\exists \mathcal{U}_0^c = \{U_{\alpha_i}^c\}_{i=1}^n$ a finite subcollection such that $\bigcap_{i=1}^n U_{\alpha_i}^c \subseteq V$. Thus $\exists U_0 = \{U_{\alpha_i}\} \ni V^c \subseteq \bigcup_{\alpha \in \lambda} U_{\alpha_i} \implies (\bigcup_{\alpha \in \lambda} U_{\alpha_i}) \cup V = X$. Since $V \in \mathcal{U}$, let $V = U_{\alpha_0}$, thus $\{U_{\alpha_i}\}_{i=0}^n$ is a finite subcover of X . Since you can construct a finite subcover of X for every cover of X , X is compact. $\square$

Claim 6.8. *Let A be a closed subspace of a compact space. Then A is compact*

Proof. Suppose A is a closed subspace of a compact space. $\forall \mathcal{U} \subseteq \mathcal{T}$, $\mathcal{U} = \{U_\alpha\}_{\alpha \in \lambda}$ is a open cover of A . For all $\alpha \in \lambda$, $\exists C_\alpha \in \mathcal{T} \ni C_\alpha \cap A = U_\alpha$, thus create $\mathcal{C} = \{C_\alpha\}_{\alpha \in \lambda}$. $A \subseteq \bigcup_{\alpha \in \lambda} C_\alpha$ thus $\mathcal{C} \cup \{A^c\}$ is a open cover of X . thus there exists a finite subcover, $\mathcal{C}_0 \cup \{A^c\}$ of X where $\mathcal{C}_0 = \{C_{\alpha_i}\}_{i=1}^n$. Thus $A \subseteq \bigcup_{i=1}^n C_{\alpha_i} \implies A \subseteq \bigcup_{i=1}^n C_{\alpha_i} \cap A \implies A \subseteq \bigcup_{i=1}^n U_{\alpha_i}$. Thus there exists a finite subcover of A , $\mathcal{U}_0 = \{U_{\alpha_i}\}_{i=1}^n$ where $U_{\alpha_i} = C_{\alpha_i} \cap A$. Thus every open cover of A has a finite subcover, thus A is compact. $\square$

Claim 6.9. *Let A be a compact subspace of a Hausdorff space X . Then A is closed.*

Proof. Suppose A is a compact subspace of a Hausdorff space X . It suffices to show A^c is open. $\forall p \in X - A, \forall a \in A$, by definition of Hausdorff space, $\exists U_a, V_a \in \mathcal{T} \ni a \in U_a, p \in V_a, U_a \cap V_a = \emptyset$. The set $\{U_a\}_{a \in A}$ is a open cover of A . Since A is compact, $\exists \{a_1, \dots, a_n\} \subseteq A$ such that $A \subseteq \bigcup_{i=1}^n U_{a_i} = U$, a finite cover. Take $V_p = \bigcap_{i=1}^n V_{a_i}$, a open set since it is a finite intersection. $p \in V$ and since $V_{a_i} \subseteq V_{a_i}^c$ for all i that means $V_p \subseteq U^c \subseteq A^c$. Thus $\forall p \in A^c, \exists V_p \in \mathcal{T} \ni p \in V_p \subseteq A^c$. So by claim 2.3, A^c is open thus A is closed. $\square$

Problem 6.10. Construct an example of a compact subset of a topological space that is not closed.

Proof. Take the two headed snake topology. $A = (0, 1] \cup \{0'\}$ is compact, $A^c = (1, \infty) \cup \{0''\}$ is not open, thus A is not closed. A is compact. $\square$

Problem 6.11. Must the intersection of two compact sets be compact? Add hypotheses, if necessary. Extend any theorems you discover, if possible.

Claim 6.11a.

Claim 6.12. Every compact, Hausdorff space is normal.

Proof. Let $A, B \in \mathcal{C} \ni A \cap B = \emptyset$. Since A and B are closed, by claim 6.9 A and B are compact. $\forall (a, b) \in A \times B$, since X is Hausdorff, $\exists U_{a,b}, V_{a,b} \in \mathcal{T} \ni a \in U_{a,b}, b \in V_{a,b}, U_{a,b} \cap V_{a,b} = \emptyset$.

$\forall a \in A$, we want to create U_a and V_a where V_a is a open cover of B and U_a is a open set which is disjoint from V_a containing a . Do this by creating the collection $\mathcal{V}_a = \{V_{a,b}\}_{b \in B}$ and $\mathcal{U}_a = \{U_{a,b}\}_{b \in B}$. $\mathcal{V}_a$ is a open cover of B . Since B is compact there must be a finite subcover of B , $\mathcal{V}_{a,0} = \{V_{a,b_i}\}_{i=1}^n$ this induces a finite subcollection of $\mathcal{U}_a = \{U_{a,b_i}\}_{i=1}^n$. Since $\forall i \in [1, \dots, n] U_{a,b_i} \subseteq V_{a,b_i}^c, \bigcap_{j=1}^n U_{a,b_j} \subseteq V_{a,b_i}^c$. Thus $\bigcap_{j=1}^n U_{a,b_j} \subseteq \bigcap_{i=1}^n V_{a,b_i}^c = (\bigcup_{i=1}^n V_{a,b_i})^c \implies \bigcup_{i=1}^n V_{a,b_i} \cap \bigcap_{j=1}^n U_{a,b_j} = \emptyset$. $\bigcap_{j=1}^n U_{a,b_j}$ is a open set, because its a finite intersection. Let $V_a = \bigcup_{i=1}^n V_{a,b_i}$ and $U_a = \bigcap_{j=1}^n U_{a,b_j}$.

Now collect all the V_a and U_a into corresponding collections, $\mathcal{U} = \{U_a\}_{a \in A}$ and $\mathcal{V} = \{V_a\}_{a \in A}$. Since $\forall a \in A a \in U_a$, $\mathcal{U}$ is a open cover of A . Since A is compact, $\exists \mathcal{U}_0 \subseteq \mathcal{U}$ where $\mathcal{U}_0 = \{U_{a_i}\}_{i=1}^m$ and $\mathcal{U}_0$ is a cover of A . This induces the subcollection of $\mathcal{V}$, $\mathcal{V}_0 = \{V_{a_i}\}_{i=1}^m$. Since $V_{a_i} \cap U_{a_i} = \emptyset$ by definition, $\forall i \in [1, \dots, m] (\bigcap_{i=1}^m V_{a_i}) \cap U_{a_i} = \emptyset \implies (\bigcap_{i=1}^m V_{a_i}) \cap (\bigcup_{i=1}^m U_{a_i}) = \emptyset$. Let $V = \bigcap_{i=1}^m V_{a_i}$ and $U = \bigcup_{i=1}^m U_{a_i}$, both are open sets because V is a finite intersection and U is a union. Since $\forall a \in A, B \subset V_a \implies B \subseteq V_{a_i}$ for all i , thus $B \subseteq V$. Thus $A \subseteq U, B \subseteq V$, and $U \cap V = \emptyset$.

Thus X is normal, by definition of normality. $\square$

Claim 6.13. Let $\mathcal{B}$ be the basis for a space X . Then X is compact if and only if every open cover of X by basis open sets in $\mathcal{B}$ has a finite subcover

Proof. $\implies$: Suppose X is compact. $\forall \mathcal{U} \subseteq \mathcal{B}$ where $\mathcal{U}$ is a cover of A . By definition of basis $\mathcal{U}$ is a open cover of A , thus there $\exists \mathcal{U}_0 \subseteq \mathcal{U}$ a finite subcover of A . $\mathcal{U}_0$ consists of basis elements. Thus the theorem is true.

$\impliedby$: Suppose every open cover of X by basis open sets in $\mathcal{B}$ has a finite subcover. Assume the contrary that X is not compact, thus $\exists \mathcal{U} = \{U_\alpha\}_{\alpha \in \lambda} \subseteq \mathcal{T}$ a cover of X which doesn't have a finite subcover. $\forall \alpha \in \lambda, \exists \{V_{\alpha,\beta}\}_{\beta \in \rho} \subseteq \mathcal{B} \ni U_\alpha = \bigcup_{\beta \in \rho} V_{\alpha,\beta}$. Thus $\{V_{\alpha,\beta}\}_{\alpha \in \lambda, \beta \in \rho}$ is a basis open set cover of A . Since $\mathcal{U}$ doesn't have a finite subcover, $\{V_{\alpha,\beta}\}_{\alpha \in \lambda, \beta \in \rho}$ cannot have a finite subcover which contradicts the assumption. Thus by contradiction X must be compact. $\square$

2 Heine Borel

Claim 6.14. *For any $a \leq b$ in $\mathbb{R}$, the subspace $[a, b]$ is compact*

Proof. $\forall \mathcal{U} \subseteq \mathcal{B} := \{(a, b) \mid a, b \in \mathbb{R}, a < b\}$, where $\mathcal{U}$ is an open cover of $[a, b]$. Let $\mathcal{U}_0 = \emptyset$, $a_0 = a$, $b_0 = b$, $I_0 = [a_0, b_0]$. Construct a subcover $\mathcal{U}_0$ of $[a, b]$ using this algorithm.

1. $i = 0$
2. while $a_i \leq b_i$:
 - (a) $\exists U, V \in \mathcal{U} \ni a_i \in U, b_i \in V$. U and V are bounded thus $\sup U, \inf U, \sup V, \inf V$ exists. Thus let U_i be the U such that $\sup U = \max \{\sup A \mid A \in \mathcal{U}, a_i \in A\}$. Let V_i be the V such that $\inf V = \min \{\inf A \mid A \in \mathcal{U}, b_i \in A\}$
 - (b) Let $a_{i+1} := \sup U_i$ and let $b_{i+1} := \sup V_i$
 - (c) $I_{i+1} := [a_{i+1}, b_{i+1}]$
 - (d) $\mathcal{U}_0 = \mathcal{U}_0 \cup \{U_i, V_i\}$
 - (e) $i = i + 1$

Now we need to prove 2 things about the algorithm:

1. It terminates
2. $\mathcal{U}_0$ is finite and a cover

Assume the contrary that the algorithm doesn't terminate, thus $\forall n \in \mathbb{N}$, $a_n \leq b_n$. Note if $a_n = b_n$ for any n , then $b_{n+1} < a_{n+1}$ because $\inf V_n < a_n < \sup U_n$. Thus it suffices to assume $\forall n \in \mathbb{N}$, $a_n < b_n$. Thus we have two sequences $\{a_n\}_{n \in \mathbb{N}}$ and $\{b_n\}_{n \in \mathbb{N}}$ where a_n is monotonically increasing and b_n is monotonically decreasing and $a \leq a_n < b_n \leq b$. Thus $a_n \rightarrow c = \sup(\{a_n\}_{n \in \mathbb{N}})$ and $b_n \rightarrow d = \inf \{b_n\}_{n \in \mathbb{N}}$ converge by MCT. Thus the interval $[c, d] \subseteq [a, b]$. Assume the contrary $\exists U \in \mathcal{U} \ni c \in U$, because there $\exists \epsilon \in \mathbb{R}^+ \ni (c - \epsilon, c + \epsilon) \subseteq U$ for this $\epsilon \exists K \in \mathbb{N} \ni \forall n \geq K, a_n \in (c - \epsilon, c + \epsilon) \implies a_n \in U$. Since $a_n < a_{n+1} < \sup U$ and $a_n \in U$, $a_{n+1} = \sup U$ which creates a contradiction. Thus $c \notin U$. Thus $\mathcal{U}$ is not an open cover of $[a, b]$ since c is not in it. Thus a contradiction occurs. Thus by contradiction the algorithm must terminate in finite iterations or i is finite.

Thus $|\mathcal{U}_0| = 2^{i+1}$ and is finite. Let f be the last i . If the algorithm terminates, it means $b_f < a_f$. The subset of $\{V_i\}_{i=0}^f \subseteq \mathcal{U}_0$ creates an open cover of $[b_f, b]$ and the subset $\{U_i\}_{i=0}^f$ is an open cover of $[a, a_f]$. Since $b_f < a_f$, $\{V_i\}_{i=0}^f \cup \{U_i\}_{i=0}^f$ is an open cover of $[a, a_f] \cup [b_f, b] = [a, b]$.

Thus for any open cover $\mathcal{U}$ of $[a, b]$, by construction there exists a finite subcover. Thus X is compact. □

Claim 6.15. (Heine Borel Theorem) *Let A be a subset of $\mathbb{R}$. Then A is compact iff A is closed and bounded*

Proof. $\implies$: By claim 6.2, A has a maximum $M \in A$ and minimum $N \in A$ where $\forall a \in A$, $N \leq a \leq M$. Thus A is bounded. $\mathbb{R}$ is a Hausdorff space, proven in chapter 4. Thus A is closed by claim 6.9.

$\impliedby$: A is closed and bounded. Since A is bounded, $\exists \sup(A), \inf(A) \in \mathbb{R}$.

Thus $A \subseteq [\sup(A), \inf(A)]$. By claim 6.14, $[\sup(A), \inf(A)]$ is compact. Since A is closed in $\mathbb{R}$ and thus closed in $[\sup(A), \inf(A)]$, by claim 6.8 it is compact. □

Problem 6.16. Consider the rationals $\mathbb{Q}$ with the subspace topology inherited from $\mathbb{R}$. Find a set A that is closed and bounded but not compact.

Proof. Let $a_n = \frac{\lfloor 10^{n+1}\pi \rfloor}{10^n}$. Let $A = \{a_n \mid n \in \mathbb{N}\}$. $\text{cl}_{\mathbb{R}}(A) = A \cup \{\pi\}$. $\text{cl}_{\mathbb{R}}(A) \cap \mathbb{Q} = A$, thus A is closed in $\mathbb{Q}$. $3 \leq a_n \leq \pi$ for all $n \in \mathbb{N}$. Thus A is bounded.

Let $\delta_n = \frac{1}{2}(a_{n+1} - a_n)$.

Take $\mathcal{U} := \{(a_n - \delta_n, a_n + \delta_n)\}_{n \in \mathbb{N}}$. $\mathcal{U}$ is a cover for A but there is no finite subcover because each open set of $\mathcal{U}$ contains 1 unique element of A . $\square$

3 Compactness and Products

Claim 6.18. (Tube Lemma): Let $X \times Y$ be a product space where Y is compact. If U is a open set containing $x_0 \times Y$, then there is a open set $W \in \mathcal{T}_x$ in X containing x_0 such that $W \times Y \subseteq U$

Proof. By claim 2.3, $\forall y \in Y, \exists V \in \mathcal{T} \ni (x_0, y) \in V \subseteq U$. By definition of product topology and basis, $\exists A_y \in \mathcal{T}_Y$ and $\exists C_y \in \mathcal{T}_X$ such that $(x_0, y) \in C_y \times A_y \subseteq V$. Create the collections using these A_y and C_y , thus $\mathcal{A} = \{A_y\}_{y \in Y}$ and $\mathcal{C} = \{C_y\}_{y \in Y}$. Since $\forall y \in Y, y \in A_y$, $\mathcal{A}$ is a open cover of Y . Since Y is compact, $\exists \mathcal{A}' = \{A_{y_i}\}_{i=1}^n$ a finite subcover of Y . This induces a finite subcollection in $\mathcal{C}$ where $\mathcal{C}_0 = \{C_{y_i}\}_{i=1}^n$. Since $\forall y \in Y, x_0 \in C_y$, thus $x_0 \in \bigcap_{i=1}^n C_{y_i} \in \mathcal{T}_X$, since its a finite intersection. Thus $W = \bigcap_{i=1}^n C_{y_i} \times \bigcup_{i=1}^n A_{y_i}$. $x_0 \times Y \subseteq W$ and since $\forall y \in Y, C_y \times A_y \subseteq U \implies \bigcap_{w \in Y} C_w \times A_y \subseteq U \implies \bigcap_{w \in Y} C_w \times \bigcap_{w \in Y} A_y \subseteq U \implies W \subseteq U$. $\square$

Claim 6.19. Let X and Y be compact spaces. Then $X \times Y$ is compact

Proof. $\forall \mathcal{U} = \{U_\alpha\}_{\alpha \in \lambda} \subseteq \mathcal{B}$ be any basis cover of $X \times Y$. Thus $\forall \alpha \in \lambda, \exists (V_\alpha, W_\alpha) \in \mathcal{T}_X \times \mathcal{T}_Y \ni U_\alpha = V_\alpha \times W_\alpha$. $\forall x \in X$ define the subcollection $\mathcal{U}_x = \{U \in \mathcal{U} \mid U \cap x \times Y \neq \emptyset\} = \{U_\alpha\}_{\alpha \in \lambda_x}$. Thus $\mathcal{U}_x$ is a open cover of $x \times Y$. Thus $\{W_\alpha\}_{\alpha \in \lambda_x}$ is a open cover of Y and thus exists a finite subcover of Y since Y is compact. Thus there exists a finite subcover of $x \times Y$, $\mathcal{U}_{x,0} = \{U_{\alpha_i}\}_{i=1}^n$. By the tube lemma, $\exists S_x \in \mathcal{T}_x \ni x \times Y \subseteq S_x \subseteq \bigcup_{i=1}^n U_{\alpha_i}$.

Now create a collection of S 's, $\mathcal{S} = \{S_x\}_{x \in X}$ is a open cover of X . Thus $\exists \mathcal{S}_0 = \{S_{x_i}\}_{i=1}^m \subseteq \mathcal{S} \ni \mathcal{S}_0$ is a finite subcover of X .

Since each $S_x \times Y$ corresponds to a finite subcover of $x \times Y$ $\mathcal{U}_{x,0}$, $\mathcal{U}_0 := \bigcup_{i=1}^m \mathcal{U}_{x_i,0}$ is finite as its the finite union of sets of finite elements.

Moreover $\forall (x, y) \in X \times Y, \exists S_{x_i} \in \mathcal{S}_0$ such that $x \in S_{x_i} \implies (x, y) \in S_{x_i} \times Y$ thus $(x, y) \in \mathcal{U}_{x_i,0} \subseteq \mathcal{U}_0$. Thus $\mathcal{U}_0$ is a cover of $X \times Y$.

Thus $\mathcal{U}_0$ is a finite subcover of $X \times Y$. Thus by construction every cover of $X \times Y$ has a finite subcover. Thus $X \times Y$ is compact by definition. $\square$

Problem 6.20. (Heine Borel PT 2) Let $A \subseteq \mathbb{R}^n$ be compact iff A is closed and bounded.

Proof. $\implies$: A is compact. By claim 6.9, A is closed since $\mathbb{R}^n$ is Hausdorff.

Assume the contrary that A is not bounded. Define $\forall x \in \mathbb{R}^n, \delta_x = d(0, x)$. Define $B(\delta_x) := \{y \in \mathbb{R}^n \mid d(0, y) < \delta_x\}$. Thus $\forall x \in A, \exists y_x \in A \ni \delta_x < \delta_{y_x} \implies y_x \notin B(\delta_{y_x})$ but $x \in B(\delta_{y_x})$. Thus create the open cover $\mathcal{U} = \{B(\delta_{y_x})\}_{x \in A}$. Any finite subcollection of $\mathcal{U}$ corresponds to $\{x_1, \dots, x_n\} \subseteq A$. $\forall n \in \mathbb{N}, \forall K = \{x_1, \dots, x_n\} \subseteq A$. Let $\mathcal{U}_K := \{B(\delta_{y_x})\}_{x \in K}$. Then the set $\delta_K = \{\delta_{x_1}, \dots, \delta_{x_n}\}$ is finite in $\mathbb{R}$ and thus $\exists m \in [1, \dots, n]$ where $\delta_{x_m} = \max \delta_K$. By definition $\exists y_m \in A \ni \delta_{y_m} > \delta_{x_m} \implies y_m \notin B(\delta_{y_m}) \implies y_m \notin \bigcup_{x \in K} B(\delta_{y_x})$. Thus $\mathcal{U}_K$ is not a cover of A . Thus a contradiction exists since A is compact. Thus A must be bounded.

$\impliedby$: A is closed and bounded. Thus $\exists r \in \mathbb{R}^+$ such that $A \subseteq B(x, r)$. $A \subseteq B(r) \subseteq [-r, r]^n$. $[-r, r]^n$ is closed as its the finite product of closed sets and compact as its the finite product of compact sets. Thus since A is closed in $\mathbb{R}^n$ its closed in $[-r, +r]^n$. Thus by 6.8, A is compact. $\square$

Claim 6.21. (Alexander Subbasis Theorem). Let $\mathcal{S}$ be a subbasis for a space X . Then X is compact if and only if every subbasic open cover has a finite subcover.

Proof. $\implies$: Suppose X is compact and suppose $\mathcal{S}$ is the subbasis collection. $\forall \mathcal{U} \subset \mathcal{S}$ where $\mathcal{U}$ is a open cover of X . Since X is compact, $\exists \mathcal{U}_0 \subseteq \mathcal{U}$ a finite subcover.

$\impliedby$: We will try to prove the contrapositive. Suppose X is not compact. Thus $\exists \mathcal{U} \subset \mathcal{T} \ni \mathcal{U}$ doesn't have a finite subcover. Let $\mathcal{U}^*$ be the set of open covers of X with no finite subcover. There exists a ordering on $\mathcal{U}^*$ of inclusion. $\square$

Problem 6.22. Use the Alexander Subbasis Theorem to prove that 2^X is compact for any X

Solution. ■

Claim 6.23. (Tychonoff's Theorem) Any product of compact spaces is compact.

Proof. $X = \prod_{\alpha \in \lambda} X_\alpha$ where X_α is compact. X has a natural subbasis of inverse projection of open sets of a given factor ie $\forall \alpha \in \lambda, \forall U \in \mathcal{T}_{X_\alpha} \pi_{X_\alpha}^{-1}(U)$. Assume the contrary X is not compact. $\exists \mathcal{U}$ a subbasis open cover of X with no finite subcover. Partition $C = \bigcup_{\alpha \in \lambda} C_\alpha$ where C_α is the subcollection of all the inverse projections of open sets of a particular factor. Being inverse projections, this means the projection of C_α onto X_α has no finite subcollection, and since each X_α is compact $\exists x_i \in X_\alpha$ which is not covered. Thus $(x_i)_\alpha \in X$ is not covered thus a contradiction exists and X must be compact. $\square$

Problem 6.24. Prove $[0, 1]^\omega$ is not compact with box topology

Proof. Assume the contrary X is compact. Let $x_n := (\delta(n, 0), \delta(n, 1), \dots)$. Let $A = \{x_n\}_{n \in \mathbb{N}}$. A is infinite and thus must have a limit point. A is closed and has no limit points. Thus by contradiction X is not compact. $\square$

Claim 6.25. *Every countable compact and Lindelof space is compact*

Proof. Open cover $\rightarrow$ (Lindelof) $\rightarrow$ countable cover $\rightarrow$ (Countable Compact) $\rightarrow$ Finite subcover $\square$

Claim 6.26. *Let X be a T_1 space. Then if X is countably compact if and only if every infinite subset of X has a limit point*

Proof. $\Rightarrow$: We will prove by contrapositive, thus suppose $\exists A \subseteq X \ni |A| \geq \aleph_0$ but A has no limit point thus A is closed and any subset of A is closed.

1. If A is countable: Since A is countable, $A = \{a_n\}_{n \in \mathbb{N}}$. Let $U_n := X - \{a_n, a_{n+1}, \dots\}$. Thus $a_n \in U_{n+1}$ for all $n \in \mathbb{N}$ and $\forall x \in X - A, x \in U_n$ for all $n \in \mathbb{N}$. But $a_n \notin U_m$ for all $n, m \in \mathbb{N} \ni m \geq n$. Thus $\{U_n\}_{n \in \mathbb{N}}$ is a countable cover of X . Any finite subcollection $\mathcal{U}_0$ corresponds to $\{n_1, \dots, n_m\} \subseteq \mathbb{N}$. $\mathcal{U}_0 = \{U_{n_i}\}_{i=1}^m$. Let $n_m = \max(\{n_i | i \in [1, \dots, m]\})$. $a_{n_m} \notin U_{n_i}$ for all i . Thus $\mathcal{U}_0$ is not a subcover of X . Thus $\mathcal{U}$ has no finite subcover. Thus X is not countably compact.
2. If A is uncountable: A has a subset B which is countable. By the condition above, $\exists x \in X$, a limit point of B and thus a limit point of A .

Thus X is not countably compact. Thus by contrapositive, if X is countably then every infinite subset of X has a limit point.

$\Leftarrow$: Suppose the 2nd half of the theorem is true. Assume the contrary that that $\exists \mathcal{U} = \{U_i\}_{i \in \mathbb{N}}$ a countable open cover with no finite subcover. Choose $x_1 \notin V_1 := U_1$. Thus $\exists n_1 \in \mathbb{N} \ni x \in U_{n_1}$. Now choose

$$x_2 \notin V_2 := \bigcup_{i=1}^{n_1} U_i$$

. Thus $\exists n_2 \in \mathbb{N} \ni x_2 \in U_{n_2}$ and note $n_2 > n_1$. Continue this process, thus

$$x_i \notin V_i := \bigcup_{i=1}^{n_{i-1}} U_i$$

but $\exists n_i \in \mathbb{N}, x_i \in U_{n_i}$ and thus $n_i > n_{i-1}$. Thus $V_{i+1} = V_{i-1} \cup U_{n_i}$.

Thus there is a sequence of $A = \{x_i | i \in \mathbb{N}\}$'s and a sequences of open sets $\mathcal{V} = \{V_i\}_{i \in \mathbb{N}}$. Thus $x_i \notin V_j$ for $i \geq j$. Since $\mathcal{U}$ is a open cover, $\{V_i\}_{i \in \mathbb{N}}$ is also a open cover of X . Since A is infinite set, $\exists x \in X$, a limit point. Since $\mathcal{V}$ is a open cover, $\exists m_0 \in \mathbb{N} \ni x \in V_{m_0}$. By the definition of limit point, $\exists m_1 \in \mathbb{N} \ni x \neq x_{m_1}, x_{m_1} \in V_{m_0} \Rightarrow m_1 < m_0$. Since X is T_1 , $\{x_{m_1}\}$ is open. Thus $x \in V_{m_0} \cap \{x_{m_1}\}^c$. By definition of limit point, $\exists m_2 \in \mathbb{N} \ni x \neq x_{m_2}, x_{m_2} \in V_{m_0} \cap \{x_{m_1}\}^c \Rightarrow x \in V_{m_0} \cap \{x_{m_1}\}^c \cap \{x_{m_2}\}^c$. One can continue this process indefinitely. However $\{1, \dots, m_0\}$ is finite. Thus a contradiction occurs as the process is both infinite and finite, thus X must be countably compact. $\square$

Claim 6.27. *If X is a Lindelof space, then every uncountable subset of X has a limit point*

Proof. Suppose $\exists A \subseteq X$ an uncountable subset has no limit point. Thus $\forall x \in X, \exists U_x \in \mathcal{T} \ni x \in U_x, (U_x - \{x\}) \cap A = \emptyset$. Thus $\mathcal{U} = \{U_x\}_{x \in X}$ is an open cover of X . Any countable subcollection of $\mathcal{U}$, corresponds to $B = \{x_i\}_{i \in \mathbb{N}} \subseteq X$ a countable subset of X . Thus $\mathcal{U}_B := \{U_{x_i}\}_{i \in \mathbb{N}}$. Since B is countable, $C = A \cap B$ must be countable and since A is uncountable $A \cap B^c$ must not be empty. Since $\forall i \in \mathbb{N} x_i \in U_{x_i}$ but $(U - \{x_i\}) \cap A = \emptyset$. Thus if $x_i \notin A \implies U_{x_i} \cap A = \emptyset$. If $x_i \in A \implies U_{x_i} \cap A = \{x_i\}$. Thus $\bigcup_{i \in \mathbb{N}} U_{x_i} \cap A = B$, and $\exists a \in A \ni a \notin \bigcup_{i \in \mathbb{N}} U_{x_i}$ since $A \cap B^c$ is uncountable. Thus $\mathcal{U}$ has no countable subcover. Thus A is not Lindelof.

Thus by contrapositive if A is Lindelof then $\nexists A \subseteq X$ uncountable with no limit. Thus all uncountable subsets of X have limit points. $\square$

Claim 6.28a. *A space X is countably compact iff every countable collection of closed sets with the finite intersection property has a nonempty intersection*

Proof. $\implies$: Suppose X is countably compact. Let $B = \{B_n\}_{n \in \mathbb{N}}$ a countable collection of closed sets with the finite intersection property. Assume the contrary that $\bigcap_{n \in \mathbb{N}} B_n = \emptyset \implies \bigcup_{n \in \mathbb{N}} B_n^c = X$. Thus $B^c = \{B_n^c\}_{n \in \mathbb{N}}$ is an open cover of X . Thus there exists a finite subcover $D^c \subseteq B^c$ since X is compact. Thus $\bigcup_{n=1}^m D_n^c = X \implies \bigcap_{n=1}^m D_n = \emptyset$ thus there exists finite subcollection $D \subseteq B$ such that the intersection of the finite subcollection is empty, thus breaking the finite intersection property of B . Thus by contradiction, B must have a nonempty intersection.

$\impliedby$: Suppose X is a topological space such that every countable collection of closed sets with the finite intersection property have a nonempty intersection. Assume the contrary that X is not compact. Thus there exists a countable open cover of X , $\mathcal{U} = \{U_\alpha\}_{\alpha \in \mathbb{N}}$ which doesn't have a finite subcover of X . $\mathcal{U}^c = \{U_\alpha^c\}_{\alpha \in \mathbb{N}}$ is a collection of closed sets. Since $\bigcup_{\alpha \in \mathbb{N}} U_\alpha = X \implies \bigcap_{\alpha \in \mathbb{N}} U_\alpha^c = \emptyset$

Thus for all finite subcollections $\mathcal{U}_0 = \{U_n\}_{n \in [0, \dots, m]}$, $\bigcup_{n=1}^m U_n \neq X \implies \bigcap_{n=1}^m U_n^c \neq \emptyset$.

Thus $\mathcal{U}^c$ has the finite intersection property, thus its intersection must be nonempty which causes a contradiction. Thus X must be countably compact by contradiction. $\square$

Claim 6.28b. *A space X is Lindelof iff every collection of closed sets with the countable intersection property has a nonempty intersection*

Proof. $\implies$: Suppose X is Lindelof. Let $B = \{B_\alpha\}_{\alpha \in \lambda}$ a collection of closed sets with the countable intersection property. Assume the contrary that $\bigcap_{\alpha \in \lambda} B_\alpha = \emptyset \implies \bigcup_{\alpha \in \lambda} B_\alpha^c = X$. Thus $B^c = \{B_\alpha^c\}_{\alpha \in \lambda}$ is an open cover of X . Thus there exists a countable subcover $D^c \subseteq B^c$ since X is compact. Thus $\bigcup_{\alpha \in \mathbb{N}} D_\alpha^c = X \implies \bigcap_{\alpha \in \mathbb{N}} D_\alpha = \emptyset$ thus there exists countable subcollection $D \subseteq B$ such that the intersection of the finite subcollection is empty, thus breaking the countable intersection property of B . Thus by contradiction, B must have a nonempty intersection.

$\impliedby$: Suppose X is a topological space such that every collection of closed sets with the countable intersection property have a nonempty intersection. Assume the contrary that X is not Lindelof. Thus there exists an open cover of X , $\mathcal{U} = \{U_\alpha\}_{\alpha \in \lambda}$ which doesn't have a countable subcover of X . $\mathcal{U}^c = \{U_\alpha^c\}_{\alpha \in \lambda}$ is a collection of closed sets. Since $\bigcup_{\alpha \in \lambda} U_\alpha = X \implies \bigcap_{\alpha \in \lambda} U_\alpha^c = \emptyset$

Thus for all countable subcollections $\mathcal{U}_0 = \{U_{\alpha_i}\}_{i \in \mathbb{N}}$, $\bigcup_{n \in \mathbb{N}} U_{\alpha_i} \neq X \implies \bigcap_{n=1}^m U_{\alpha_i}^c \neq \emptyset$.

Thus $\mathcal{U}^c$ has the countable intersection property, thus its intersection must be nonempty which causes a contradiction. Thus X must be Lindelof by contradiction. $\square$

Claim 6.28c. *A space X is countably compact if for any open set $U \subset X$ and any countable collection of closed sets such that $\bigcap_{n \in \mathbb{N}} K_n \subseteq U$, there is a finite number of K_n whose intersection lies in U*

Proof. Simply transform proof for 6.4. Not too hard. $\square$

Claim 6.28d. *A space X is Lindelof if for any open set $U \subset X$ and any collection of closed sets such that $\bigcap_{n \in \mathbb{N}} K_n \subseteq U$, there is a countable subcollection of K_n whose intersection lies in U*

Proof. Simply transform proof for 6.4. Not too hard. $\square$

Claim 6.29a. *If A is a closed subspace of a countably compact space X , then A is countably compact*

Proof. Let $\mathcal{U} = \{U_n\}_{n \in \mathbb{N}}$ be any countable open cover of A . Since A is closed, A^c is open. Thus $\mathcal{U} \cup \{A^c\}$ is a open cover of X . Since X is countably compact, $\exists \mathcal{U}_0 = \{U_{n_i}\}_{i=1}^m$ a finite subset of $\mathcal{U}$ such that $\mathcal{U}_0 \cup \{A^c\}$ is a finite open cover of X . $\forall a \in A$, $a \in \bigcup_{i=1}^m U_{n_i}$. Thus $\mathcal{U}_0$ is a finite open cover of A . Thus by construction any countable open cover of A has a finite open cover. Thus A is countably compact. $\square$

Claim 6.29b. *If A is a closed subspace of a Lindelof space X , then A is Lindelof*

Proof. Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in \lambda}$ be any open cover of A . Since A is closed, A^c is open. Thus $\mathcal{U} \cup \{A^c\}$ is a open cover of X . Since X is Lindelof, $\exists \mathcal{U}_0 = \{U_{\alpha_i}\}_{i \in \mathbb{N}}$ a countable subset of $\mathcal{U}$ such that $\mathcal{U}_0 \cup \{A^c\}$ is a countable open cover of X . $\forall a \in A$, $a \in \bigcup_{i \in \mathbb{N}} U_{\alpha_i}$. Thus $\mathcal{U}_0$ is a countable open cover of A . Thus by construction any open cover of A has a countable open cover. Thus A is Lindelof. $\square$

Claim 6.30. *Every regular, Lindelof space is normal*

Proof. For all disjoint closed sets A and B we want to show $\exists C, D \in \mathcal{T} \ni A \subseteq C, B \subseteq D, C \cap D = \emptyset$.

By definition of regular spaces, $\forall a \in A, \exists U_a, V_a \in \mathcal{T} \ni a \in U_a, B \subseteq V_a, U_a \cap V_a = \emptyset$. Similarly, $\forall b \in B, \exists U_b, V_b \in \mathcal{T} \ni b \in U_b, A \subseteq V_b, U_b \cap V_b = \emptyset$. Thus create the collections $U = \{U_a\}_{a \in A}$ and $V = \{V_b\}_{b \in B}$. Since A and B are closed, they are Lindelof (Claim 6.29b). Thus $\exists U_0 = \{U_{a_i}\}_{i \in \mathbb{N}} \subseteq U$ a countable subcover of A . Similarly $\exists V_0 = \{V_{b_i}\}_{i \in \mathbb{N}} \subseteq V$, a countable subcover of B .

Assume the contrary that $\exists i \in \mathbb{N}$ where $\exists x \in B$ a limit point of U_{a_i} . Then $V_{a_i} \cap U_{a_i}$ cannot be empty thus a contradiction occurs and by contradiction no limit point of U_{a_i} for all i is in B . Thus $\overline{U_{a_i}} \cap B = \emptyset$.

Assume the contrary that $\exists i \in \mathbb{N}$ where $\exists x \in A$ a limit point of V_{b_i} . Then $V_{b_i} \cap U_{b_i}$ cannot be empty thus a contradiction occurs and by contradiction no limit point of U_i for all i is in B . Thus $\overline{V_{b_i}} \cap A = \emptyset$.

Thus apply normality lemma, $\exists C, D \in \mathcal{T} \ni A \subseteq C, B \subseteq D, C \cap D = \emptyset$.

Thus X is normal. $\square$

Claim 6.31. *Let $\mathcal{B}$ be a basis of X . If X is Lindelof iff any basis cover of X has a countable subcover*

Proof. Nearly same proof as 6.13, but replace finite with countable $\square$

Claim 6.32. *Every 2nd countable space is Lindelof.*

Proof. 2nd countable $\implies$ countable basis $\implies$ any basis cover is countable thus automatically Lindelof $\square$

Paracompactness

Claim 6.38. *Let $\mathcal{B} = \{B_\alpha\}_{\alpha \in \lambda}$ be a locally finite collection of subsets of the space X . Then*

$$\text{cl} \left(\bigcup_{\alpha \in \lambda} B_\alpha \right) = \bigcup_{\alpha \in \lambda} \text{cl}(B_\alpha)$$

Proof. One direction of this equality is obvious, $\bigcup_{\alpha \in \lambda} \text{cl}(B_\alpha) \subseteq \text{cl}(\bigcup_{\alpha \in \lambda} B_\alpha)$. It suffices to prove the other direction.

$\forall x \in X \setminus \bigcup_{\alpha \in \lambda} \text{cl}(B_\alpha)$. By the locally finiteness, $\exists U \in \mathcal{T} \ni x \in U$ but U only intersects finitely many B_α 's, thus label these sets $B_1, \dots, B_m$. Let $U' = U \cap (\bigcap_{i=1}^m X - \text{cl}(B_i))$ be another open set. $p \in U'$ but $U' \cap S_i = \emptyset \implies U' \cap \bigcup_{\alpha \in \lambda} B_\alpha = \emptyset$ thus x is not a limit point. Thus, $\bigcup_{\alpha \in \lambda} \text{cl}(B_\alpha)$ is closed. Thus by definition of closure, $\bigcup_{\alpha \in \lambda} B_\alpha \subseteq \bigcup_{\alpha \in \lambda} \text{cl}(B_\alpha) \implies \text{cl}(\bigcup_{\alpha \in \lambda} B_\alpha) \subseteq \bigcup_{\alpha \in \lambda} \text{cl}(B_\alpha)$.

Thus equality holds. $\square$

Claim 6.39. *Let A be a closed subspace of a paracompact space. Then A is paracompact.*

Proof. A is Hausdorff since Hausdorff is inherited.

Let $\mathcal{U}$ be a open cover of A . Thus $\mathcal{U}' = \mathcal{U} \cup \{A^c\}$ is a open cover of X . By paracompactness of X , there $\exists \mathcal{V}'$ a locally finite open refinement of $\mathcal{U}'$. You can partition $\mathcal{V}'$ into the open sets which are subsets of A^c and open sets which are subsets of elements of $\mathcal{U}$ to make the collections $\mathcal{V}$ and $\mathcal{W}$ respectively. Note if there is a element of the collection which is in the intersection of A^c and a element of $\mathcal{U}$ put it in the collection $\mathcal{W}$, or the open cover of A . Thus $\mathcal{V}' = \mathcal{V} \cup \mathcal{W}$. $\mathcal{W}$ is a open refinement of $\mathcal{U}$ and a open cover of A and is also locally finite since $\mathcal{V}'$ is locally finite. $\square$

Claim 6.40. *Every paracompact space is normal*

Proof. Let us first prove X , a paracompact space is regular.

Let $A \neq \emptyset, X$ be a closed set of X and $\forall x \in X \setminus A$. By the Hausdorff property, $\forall a \in A, \exists U_a, V_a \in \mathcal{T} \ni x \in U_a, a \in V_a, U_a \cap V_a = \emptyset$. $U_a \subseteq X - V_a \implies \overline{U_a} \subseteq X - V_a$ and vice versa for V_a . Thus $\mathcal{V} = \{V_a\}_{a \in A}$ is a open cover of A . Since X is paracompact and A is closed, by 6.39 A is paracompact. Thus there is a locally finite open refinement of $\mathcal{V}$, $\mathcal{W} = \{W_\alpha\}_{\alpha \in \lambda}$. Note $\forall \alpha \in \lambda, \exists a \in A$ such that $W_\alpha \subseteq V_a \implies \overline{W_\alpha} \cap U_a = \emptyset \implies x \notin \overline{W_\alpha}$. Thus $a \notin \bigcup_{\alpha \in \lambda} \overline{W_\alpha} = \bigcup_{\alpha \in \lambda} \overline{W_\alpha}$ (Claim 6.38). Thus let $U = X - \bigcup_{\alpha \in \lambda} \overline{W_\alpha}$ and $V = \bigcup_{\alpha \in \lambda} W_\alpha$ and $x \in U, A \subseteq V$ and $U \cap V = \emptyset$. Thus X is regular.

Now using regularity lets prove normality. Let A and B be disjoint closed sets. $\forall a \in A$, by regularity $\exists U_a, V_a \in \mathcal{T} \ni a \in U_a, B \subseteq V_a, U_a \cap V_a = \emptyset$. As proven above, $B \cap \text{cl}(U_a) = \emptyset$ and $a \notin \text{cl}(V_a)$. Since A is closed, A is paracompact. Thus $\mathcal{U} = \{U_a\}_{a \in A}$ is a open cover of A . By paracompactness, $\exists \mathcal{D} = \{D_\alpha\}_{\alpha \in \lambda}$ a locally finite open refinement of $\mathcal{U}$. Note $\forall \alpha \in \lambda, \exists a \in A$ such that $D_\alpha \subseteq U_a \implies a \notin \text{cl}(D_\alpha)$. Thus $B \cap \text{cl}(\bigcup_{\alpha \in \lambda} D_\alpha) = \emptyset$ (claim 3.38). Thus $U = X - \text{cl}(\bigcup_{\alpha \in \lambda} D_\alpha)$ and $V = \bigcup_{\alpha \in \lambda} D_\alpha$. Thus X is normal. $\square$

Claim 6.41. *Every regular, T_1 , Lindelof space is paracompact*

Proof. Let $\mathcal{U}$ be open cover of X , WLOG it is countable since X is Lindelof. $\forall x \in X, \exists U_x \in \mathcal{U} \ni x \in U_x$. Thus for x and $X - U_x$, $\exists V_x, W_x \in \mathcal{T} \ni x \in V_x, X - U_x \subseteq W_x \ni V_x \cap W_x = \emptyset \implies V_x \subseteq U_x$. Thus for all $U \in \mathcal{U}$ create $\mathcal{U}_x$ by intersecting all U with W_x if $U \neq U_x$ and preserving U_x . $\mathcal{U}_x$ is a refinement where for all $x \in U_x$, U_x only intersects with itself in $\mathcal{U}_x$. Do this for all X to get $\mathcal{U}_f$. Thus each element of $\mathcal{U}_f$ only intersects with itself while each step is a refinement of the previous step. Thus $\mathcal{U}_f$ is a locally fine refinement. $\square$